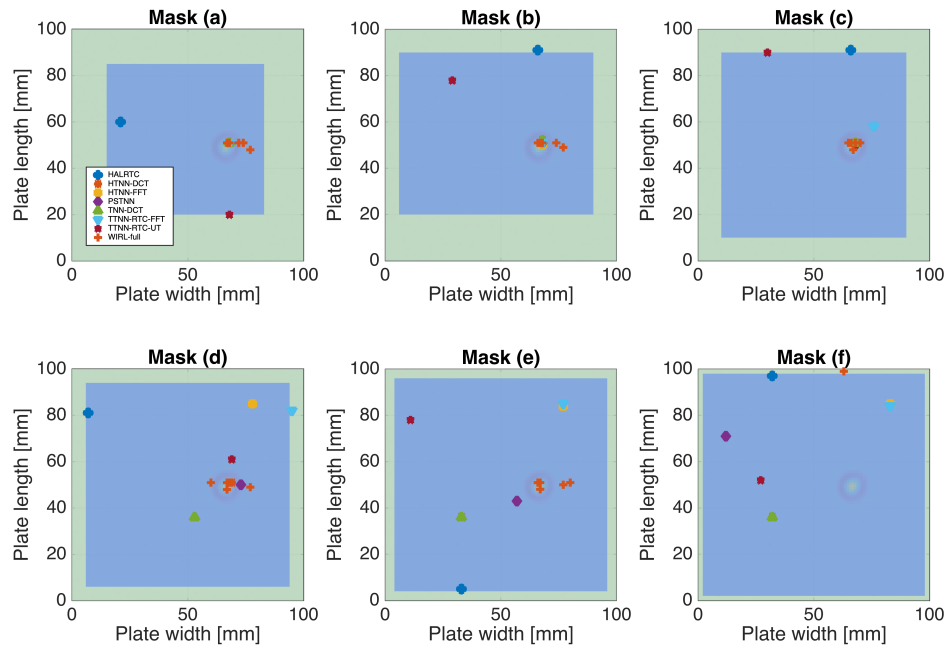


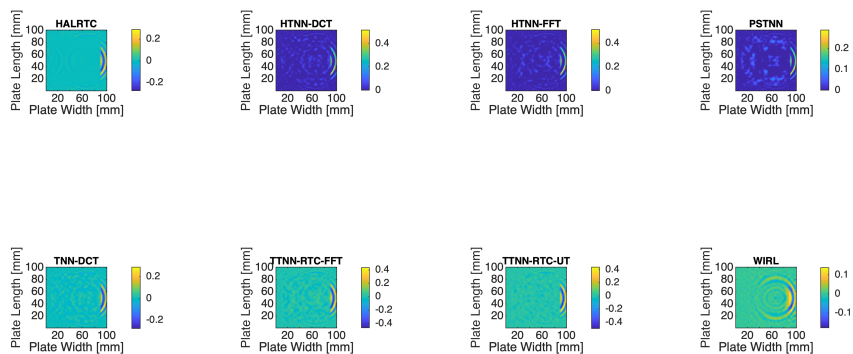
Preprocess data and mask to have the required shape for further steps

Computation of Eigenvalues and Eigenvectors of Second Derivative Matrix to use for reconstruction

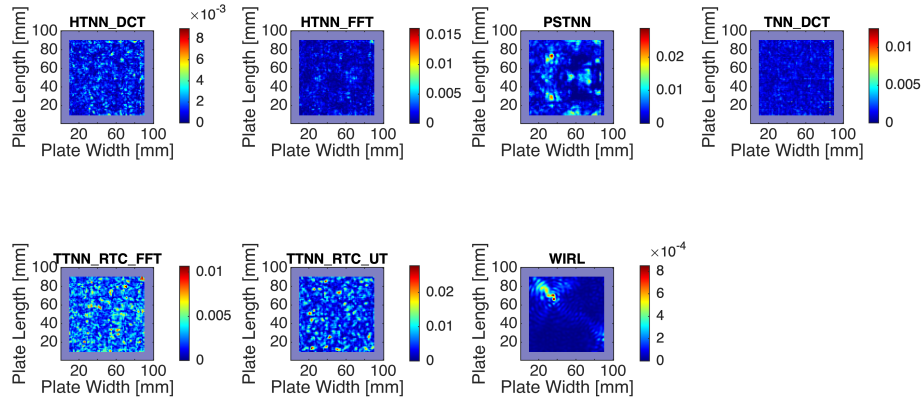
Primary Source Localization



Reconstructions: index = 3, slice 10



Defect Detection: slice = 3



Primary Source Positions by Algorithm and Mask:

	(a)	(b)	(c)	(d)	(e)	(f)
HALRTC	{ '[59,20]' }	{ '[90,65]' }	{ '[90,65]' }	{ '[80,6]' }	{ '[4,32]' }	{ '[96,31]' }
HTNN_DCT	{ '[50,67]' }	{ '[51,67]' }	{ '[50,67]' }	{ '[50,68]' }	{ '[35,32]' }	{ '[35,31]' }
HTNN_FFT	{ '[50,67]' }	{ '[49,67]' }	{ '[49,67]' }	{ '[84,77]' }	{ '[83,76]' }	{ '[84,82]' }
PSTNN	{ '[50,67]' }	{ '[50,67]' }	{ '[49,67]' }	{ '[49,72]' }	{ '[42,56]' }	{ '[70,11]' }
TNN_DCT	{ '[50,67]' }	{ '[51,67]' }	{ '[50,67]' }	{ '[35,52]' }	{ '[35,32]' }	{ '[35,31]' }
TTNN_RTC_FFT	{ '[50,67]' }	{ '[50,67]' }	{ '[57,75]' }	{ '[81,94]' }	{ '[84,76]' }	{ '[83,82]' }
TTNN_RTC_UT	{ '[19,67]' }	{ '[77,28]' }	{ '[89,29]' }	{ '[60,68]' }	{ '[77,10]' }	{ '[51,26]' }
WIRL-1	{ '[50,66]' }	{ '[50,66]' }	{ '[50,65]' }	{ '[50,66]' }	{ '[50,65]' }	{ '[98,62]' }
WIRL-2	{ '[50,73]' }	{ '[50,73]' }	{ '[47,66]' }	{ '[47,66]' }	{ '[49,76]' }	{ '[98,62]' }
WIRL-3	{ '[50,67]' }	{ '[50,67]' }	{ '[50,67]' }	{ '[48,76]' }	{ '[47,66]' }	{ 'N/A' }
WIRL-4	{ '[47,76]' }	{ '[48,76]' }	{ '[50,69]' }	{ '[50,67]' }	{ '[50,66]' }	{ 'N/A' }
WIRL-full	{ '[50,71]' }	{ '[50,65]' }	{ '[50,64]' }	{ '[50,59]' }	{ '[50,79]' }	{ 'N/A' }

Position Error Norms (distance from true position):

True Position: [49, 66]

	(a)	(b)	(c)	(d)	(e)	(f)
HALRTC	47.074	41.012	41.012	67.535	56.4	58.6
HTNN_DCT	1.4142	2.2361	1.4142	2.2361	36.77	37.696
HTNN_FFT	1.4142	1	1	36.688	35.44	38.484
PSTNN	1.4142	1.4142	1	6	12.207	58.873
TNN_DCT	1.4142	2.2361	1.4142	19.799	36.77	37.696
TTNN_RTC_FFT	1.4142	1.4142	12.042	42.521	36.401	37.577
TTNN_RTC_UT	30.017	47.202	54.489	11.18	62.61	40.05
WIRL-1	1	1	1.4142	1	1.4142	49.163
WIRL-2	7.0711	7.0711	2	2	10	49.163
WIRL-3	1.4142	1.4142	1.4142	10.05	2	NaN

WIRL-4	10.198	10.05	3.1623	1.4142	1	NaN
WIRL-full	5.099	1.4142	2.2361	7.0711	13.038	NaN

Position Error Norms (distance from true defect position):

True Defect Position: [49, 66]

	(a)	(b)	(c)	(d)	(e)	(f)
HTNN_DCT	69.814	33.302	77.104	77.104	56.436	74.411
HTNN_FFT	57.28	56.321	56.321	56.321	56.321	71.84
PSTNN	38.484	8.544	8.9443	8.0623	77.104	57.974
TNN_DCT	13.454	56.321	56.321	56.321	56.321	73.756
TTNN_RTC_FFT	37.336	43.324	43.324	64.938	41.11	32.062
TTNN_RTC_UT	39.812	51.923	53.46	82.073	68.622	54.489
WIRL	6.0828	7	6	7.2111	6	77.414

PSNR Values for each algorithm

PSNR Results:

	Mask (a)	Mask (b)	Mask (c)	Mask (d)	Mask (e)	Mask (f)
HALRTC	{[30.6458]}	{[29.4733]}	{[29.1987]}	{[28.4702]}	{[28.1654]}	{[27.8740]}
HTNN_DCT	{[28.1763]}	{[28.0651]}	{[28.0227]}	{[27.7654]}	{[27.5903]}	{[27.6212]}
HTNN_FFT	{[28.1000]}	{[28.0121]}	{[27.9691]}	{[27.7427]}	{[27.5719]}	{[27.6097]}
PSTNN	{[28.8083]}	{[28.4389]}	{[28.3307]}	{[27.9491]}	{[27.9045]}	{[27.7483]}
TNN_DCT	{[31.0365]}	{[29.7134]}	{[29.4247]}	{[28.5926]}	{[28.2497]}	{[27.9115]}
TTNN_RTC_FFT	{[28.8177]}	{[28.5766]}	{[28.4694]}	{[27.9309]}	{[27.5022]}	{[27.5976]}
TTNN_RTC_UT	{[28.4211]}	{[28.3086]}	{[28.1998]}	{[27.7671]}	{[27.3737]}	{[27.5345]}
WIMF_recon	{[29.8060]}	{[29.0941]}	{[28.9697]}	{[28.3495]}	{[28.0279]}	{[27.6435]}